

CASE 1, TREATMENT 1

Patient is a 68 yo male outpatient with history of mania and chronic pain, current cannabis use and mild neurocognitive disorder with behavioral issues (but no benzos or other seizure medications or delirium). Presenting for first treatment feeling uncertain.

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (yes)	Dx bipolarity or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Dx chronic pain
binary (yes)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

For above 3 “yes responses”

If y then -->	Consider methohexital and precedex
If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
If y then -->	Start RUL, prioritize Precedex or sufentanyl and Avoid propofol, sevoflurane, isoflurane, and midazolam

Medical Factors

68 Age

57 resting HR

125 systolic

75 diastolic

82.5kg weight

list of medications: Latuda, liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications (Skip if 1st ever treatment)

Tampa Site A Group (Sullivan, Stewart, Kroger, Gunther) algorithm suggests: Methohexital because methohexital and precede as per above have been identified as the ideal agents from our formulary. As precedex cannot be used alone and we will initially trial monotherapy our answer will be methohexital. Tylenol will be provided orally post treatment as needed due to the chronic pain factor and the patient will start RUL with stimulus based on age 68%. Dosing calculation for methohexital is as follows for the above kg.

IV Dosing of ECT anesthetics	minimum	maximum
methohexital (1-1.5mg/kg)	82.5	123.8

Cocktail: Tylenol+ precedex+ methohexital + succinylcholine

Complication: None

CASE 1, TREATMENT 2

Patient is a 68 yo male outpatient with history of mania and chronic pain, current cannabis use and high-risk behaviors with cognitive issues (but no benzos or other seizure medications or delirium) with a resting HR 57 and BP 125/75 with no known arrhythmias weighing 82.5kg. Presenting for second treatment after experiencing reemergence delirium and muscle soreness that resulted in disturbance in gait.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

82.5kg weight

list of medications: Latuda, liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications:

Input types	Subsequent treatment Factors
binary (yes)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

If y then -->	Start RUL, prioritize Precedex or sufentanyl and Avoid propofol, sevoflurane, isoflurane, and midazolam
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Cocktail: Tylenol+ precedex+ methohexital + succinylcholine

Complication: None

CASE 1, TREATMENT 3

Patient is a 68 yo male outpatient with history of mania and chronic pain, current cannabis use and high-risk behaviors with cognitive issues (but no benzos or other seizure medications or delirium) with a resting HR 57 and BP 125/75 with no known arrhythmias weighing 82.5kg. Presenting for second treatment after experiencing reemergence delirium and muscle soreness that resulted in disturbance in gait.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

82.5kg weight

list of medications: Latuda, liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Rationale: no complication no change

Cocktail: Tylenol+ precedex+ methohexital + succinylcholine

Complication: None

CASE 1, TREATMENT 4

Patient is a 68 yo male outpatient with history of mania and chronic pain, current cannabis use and high-risk behaviors with cognitive issues (but no benzos or other seizure medications or delirium) with a resting HR 57 and BP 125/75 with no known arrhythmias weighing 82.5kg. Presenting for second treatment after experiencing reemergence delirium and muscle soreness that resulted in disturbance in gait.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

82.5kg weight

list of medications: liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Rationale: no complication no change

Cocktail: Tylenol+ precedex+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 1

Patient is a 38 yo male outpatient with history of MDD, severe with PTSD + comorbid substance use (alcohol, cocaine and cannabis), severe ADHD, and chronic pain. Current cannabis use and severe alcohol use but no benzos or other seizure medications or delirium) but no known arrhythmias weighing 90kg. Presenting now for first treatment.

Input types	Psychiatric factors most important for the patient case
binary (no)	prior reemergence or hx violence
binary (no)	benzos
binary (no)	on seizure medications
binary (yes)	current pain
binary (no)	current nausea or emesis
binary (yes)	logistics w. high risk for cognitive issues

If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
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If y then -->	Start RUL, prioritize Precedex or sufentanyl and Avoid propofol, sevoflurane, isoflurane, ketamine, and midazolam
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Medical Factors

38 Age

77 HR

145 systolic 90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications (Skip if 1st ever treatment)

Tampa Site A Group (Sullivan, Stewart, Kroger, Gunther) algorithm suggests: Methohexital or propofol with a premedication. Between the two anesthetics, propofol based on anxiety and concurrent alcohol use was anticipated to most likely to wake the patient up smoother..

Tylenol will be provided orally post treatment as needed due to the chronic pain factor and the patient will start RUL with stimulus based on age 38%. After discussion of plan with patient he disclosed doubts regarding treatment response and therefore a decision was made to start with bitemporal and titrate laterality after gauging cognitive adverse effects. Dosing calculation for propofol is as follows for the above kg.

Cocktail: Tylenol+ propofol + succinylcholine

Complication: Headache, HTN

CASE 2, TREATMENT 2

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD. Consult for sleep clinic for sleep apnea evaluation was also placed.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (yes)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ propofol + succinylcholine + labetalol

Complication: None

CASE 2, TREATMENT 3

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Previous treatment canceled due to interim cocaine use. Patient is now drug free for 2 weeks with negative UDS.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Cocktail: Tylenol+ Toradol+ propofol + succinylcholine + labetalol

Complication: None

CASE 2, TREATMENT 4

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Patient started wearing CPAP machine.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Cocktail: Tylenol+ Toradol+ propofol + succinylcholine + labetalol

Complication: None

CASE 2, TREATMENT 5

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

135 systolic

80 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Cocktail: Tylenol+ Toradol+ propofol + succinylcholine

Complication: None

CASE 2, TREATMENT 6

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

70 resting HR

130 systolic

85 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Cocktail: Tylenol+ Toradol+ propofol + succinylcholine

Complication: None

CASE 2, TREATMENT 7

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

70 resting HR

130 systolic

85 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Cocktail: Tylenol+ Toradol+ propofol + succinylcholine

Complication: Inadequate seizure

CASE 2, TREATMENT 8

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

If y then -->	Change to etomidate (or methohexital if on propofol), Increase power, RUL-> BL, add stimulants (bupropion, caffeine, theophylline), increase hyperventilation
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Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 9

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 10

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 11

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 12

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 13

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 14

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 15

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs)
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Rationale: Although generally managed by PCP, severe perioperative HTN may result in excess periprocedural bleeding, myocardial ischemia and/or infarction, congestive heart failure (CHF) and acute pulmonary edema (APE). Therefore it is crucial to apply the evidence to stratify the commonly used anesthetics for ECT in order of most to least likely to contribute to hypertensive crisis vs cause one.

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 2, TREATMENT 16

Patient is a 38 yo male outpatient with history of MDD, moderately severe with PTSD complicated by comorbid substance use (alcohol, cocaine and cannabis) and chronic pain. Patient presents to ECT clinic with current cannabis use and recent alcohol use and has a history of cocaine use in the past year. Presenting for second treatment after experiencing hypertensive urgency and muscle soreness that resulted in worsening of chronic pain and associated PTSD.

Psychiatric and genetic factors no changes

Medical Factors no changes

38 Age

77 resting HR

145 systolic

90 diastolic

90 kg weight

list of medications: none

0 Current unstable fracture

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

Cocktail: Tylenol+ Toradol+ methohexital + succinylcholine

Complication: None

CASE 3, TREATMENT 1

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with a resting HR 73 and BP 135/80 with no known arrhythmias weighing 82.5kg. Presenting for first treatment.

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Dx bipolarity or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Current chronic pain
binary (no)	Dx neurocognitive disorder
Binary (yes)	Dx fracture or neuromuscular disorder

For above 3 “yes responses”

If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
If y then -->	Use rocuronium for paralytic; Avoid succinylcholine ; IF ROC THEN ALSO SUGAMMADEX

Medical Factors

68 Age

57 resting HR

125 systolic 75 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

0 Current unstable fracture

Prior Complications (Skip if 1st ever treatment)

Tampa Site A Group (Sullivan, Stewart, Kroger, Gunther) algorithm suggests: Methohexital because methohexital and precedex as per above have been identified as the ideal agents from our formulary. As precedex cannot be used alone and we will initially trial monotherapy our answer will be methohexital. Tylenol will be provided orally post treatment as needed due to the chronic pain factor and the patient will start RUL with stimulus based on age 68%. Dosing calculation for methohexital is as follows for the above kg.

IV Dosing of ECT anesthetics

	minimum	maximum
methohexital (1-1.5mg/kg)	82.5	123.8

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: Headache, nausea

CASE 3, TREATMENT 2

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 3

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 4

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 5

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 6

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 7

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 8

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 3, TREATMENT 9

Patient is a 38 yo female outpatient with history of mania and chronic pain and medical comorbidity of myasthenia gravis, ocular syndrome with no known arrhythmias weighing 85kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

66 resting HR

125 systolic

82 diastolic

85kg weight

list of medications: acyclovir, rizatriptan, acetazolamide

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium
If y then -->	Add zofran

Rationale: Add supportive meds for nausea and headache

Cocktail: Tylenol+ Methohexital + rocuronium + sugammadex

Complication: No complications

CASE 4, TREATMENT 1

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg. Presenting for first treatment after not having taken morning BP medications as requested.

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Dx bipolarity or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (no)	Dx chronic pain
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

Since patient forgot to take blood pressure medications: out of an abundance of caution,

Systolic if high then-->	avoid ketamine and consider propofol and fentanyl
If diastolic high then-->	avoid ketamine and consider propofol and fentanyl

Since fentanyl alone cannot be used, we must use propofol and consider using fentanyl to spare effect on seizure threshold.

Medical Factors

60 Age

65 resting HR

170 systolic

85 diastolic

82.2kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Tampa Site A Group (Sullivan, Stewart, Kroger, Gunther) algorithm suggests: Propofol 150mg +/- fentanyl Since none of the above conditions met, Default Path is methohexital, propofol, or etomidate, given concern for hypertension and the patient being on bupropion, the choice was made to proceed with propofol with the thought that it was less likely to result in an effect on seizure threshold as bupropion may potentially decrease seizure threshold (though still limited evidence for that claim). Starting Right unilateral given hypertensive cerebrovascular disease evident on MRI.

Cocktail: propofol + succinylcholine

Complicated by inadequate seizure

CASE 4, TREATMENT 2

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg. Presenting for first treatment after having taken morning BP medications as requested. Reports very poor sleep and headache, admitting that he has a history of headaches that do tend to respond to Tylenol.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (Yes)	Inadequate seizure

If y then -->	Change to methohexital, etomidate, add stimulants, add caffeine, theophylline)
---------------	--

Cocktail: Tylenol+ methohexital + succinylcholine

Complicated by inadequate seizure, but immediate change from RUL to bilateral with good response.

CASE 4, TREATMENT 3

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (Yes)	Inadequate seizure

If y then -->	Change to bilateral and increase power , change to etomidate, add stimulants, add caffeine, theophylline)
---------------	--

Cocktail: Tylenol+ methohexital + succinylcholine

Complication: insufficient seizure

CASE 4, TREATMENT 4

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure

If y then -->	Etomidate add stimulants (caffeine not in formulary , oral theophylline)
---------------	---

Cocktail: Tylenol+ etomidate + succinylcholine

Complication: yelling after treatment (reemergence delirium) and HTN emergency

CASE 4, TREATMENT 5

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (yes)	Reemergence Delirium
binary (yes)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

If y then - **prioritize Precedex** or sufentanyl (no RUL because of prior inadequate seizures, -> patient not on any meds to avoid

Precedex does not affect seizure threshold and therefore is favored over propofol

If y then --> **Give BP meds**, avoid ketamine and consider propofol and fentanyl

Cocktail: Tylenol+ etomidate + precedex+ succinylcholine

No complications

CASE 4, TREATMENT 6

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Rationale: No complications, no changes

Cocktail: Tylenol+ etomidate + precedex+ succinylcholine

No complications

CASE 4, TREATMENT 7

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Rationale: No complications, no changes

Cocktail: Tylenol+ etomidate + precedex+ succinylcholine

No complications

CASE 4, TREATMENT 8

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Rationale: No complications, no changes but increasing power.

Cocktail: Tylenol+ etomidate + precedex+ succinylcholine

No complications

CASE 4, TREATMENT 9

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D with a resting HR 88 and BP 115/71 with no known arrhythmias weighing 82.2kg. Premedicated with theophylline.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Tylenol+ etomidate + precedex+ succinylcholine

Complicated by insufficient seizure, prescribed theophylline to take night prior to subsequent session.

CASE 4, TREATMENT 10

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D, paced with no known arrhythmias weighing 82.2kg. Premedicated with theophylline. Reports feeling a lot better than when he first started treatment and notes that he has been able to get his car out of his garage and work on it.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

140 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure

Cocktail: Premedicated with theophylline + etomidate and precedex + labetalol

No complications.

CASE 4, TREATMENT 11

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D with a resting HR 88 and BP 115/71 with no known arrhythmias weighing 82.2kg. Increased theophylline taken.

Psychiatric and genetic factors no changes

Medical Factors no changes

65 resting HR

130 systolic

70 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Increased theophylline + etomidate and precedex + labetalol

No complications.

CASE 4, TREATMENT 12

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D with a resting HR 88 and BP 115/71 with no known arrhythmias weighing 82.2kg. Premedicated with theophylline.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Premedicated with theophylline + etomidate and precedex + labetalol

No complications.

CASE 4, TREATMENT 13

Patient is a 60 yo black male outpatient with history of MDD, anxiety, and NICMO 2/2 HTN s/p CRT-D with a resting HR 88 and BP 115/71 with no known arrhythmias weighing 82.2kg. Premedicated with theophylline.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

82.5kg weight

list of medications: atorvastatin, lithium, mirabegron, bupropion, mirtazapine, metoprolol, Entresto, omeprazole, tamsulosin

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Premedicated with theophylline + etomidate and precedex + labetalol

No complications.

CASE 5, TREATMENT 1

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. Presenting for first treatment with DOP negative pregnancy test.

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Dx bipolarity or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Dx chronic pain: headache
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

By default: methohexital

If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
---------------	--

Medical Factors

32 Age

62 resting HR

110 systolic

65 diastolic

62kg weight

list of medications: escitalopram

Prior Complications (Skip if 1st ever treatment)

IV Dosing of ECT anesthetics	minimum	maximum
methohexital (1-1.5mg/kg)	62	93

Cocktail: Tylenol+ methohexital + succinylcholine

Complicated: by hypotensive shock this treatment requiring phenylephrine.

CASE 5, TREATMENT 2

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. Presenting for second treatment after experiencing hypotension requiring phenylephrine.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

62kg weight

list of medications: escitalopram

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (yes)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

If y then -- >	Add ketamine , Avoid anesthetics that have hypotension as a side effect and possibly provide ideal pressor for patient (based on pharmacological factors)
----------------------	--

Other consideration: glycopyrrolate can also be considered if only modest increases in blood pressure are needed or if we need to set a floor for the heart rate without increasing risk of agitation. In this case, the significant drop in

Cocktail: Tylenol+ methohexital + ketamine + succinylcholine

No complications

CASE 5, TREATMENT 3

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. Presenting for third treatment after no complications on second. She notes she has been able to eat a bit more for breakfast and sleep has been a bit longer but she still awakens very early. Primary psychiatrist transitioned escitalopram to venlafaxine.

Psychiatric and genetic factors no changes

Medical Factors no changes

58 resting HR

105systolic

67 diastolic

62kg weight

list of medications: venlafaxine

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Tylenol+ methohexital + ketamine + succinylcholine

No complications

CASE 5, TREATMENT 4

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg.

Psychiatric and genetic factors no changes

Medical Factors no changes

55 resting HR

100 systolic

67 diastolic

62kg weight

list of medications: venlafaxine

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Tylenol+ methohexital + ketamine + succinylcholine

No complications

CASE 5, TREATMENT 5

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. No complications in prior treatment and she is currently being treated with methohexital and ketamine with Tylenol premedication and denies cognitive dulling or disorientation.

Psychiatric and genetic factors no changes

Medical Factors no changes

60 resting HR

110 systolic

70 diastolic

63kg weight

list of medications: venlafaxine

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Tylenol+ methohexital + ketamine + succinylcholine

No complications. (transitioned from right unilateral to bilateral during this treatment to elicit)

CASE 5, TREATMENT 6

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. No complications in prior treatment and she is currently being treated with methohexital and ketamine with Tylenol premedication and she has been tolerating switch to bilateral for inadequate seizure at the last treatment. Thus far patient has experienced drastic improvement with affective stability persisting throughout the day.

Psychiatric and genetic factors no changes

Medical Factors no changes

60 resting HR

110 systolic

70 diastolic

64kg weight

list of medications: venlafaxine + addition of bupropion

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

If y then -->

Change to etomidate, ~~Increase power, RUL->BL~~, add stimulants (**bupropion**, caffeine, theophylline), continue aggressive hyperventilation

Cocktail: Tylenol+ etomidate + ketamine + succinylcholine

Complicated by hypertensive emergency level BP and nausea

CASE 5, TREATMENT 7

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. Since initiation of treatment, patient has resumed PO intake, changed psychiatric medications in favor of sympathomimetic effects and has experienced improvement in energy.

Psychiatric and genetic factors no changes

Medical Factors no changes

62 resting HR

110 systolic

65 diastolic

65kg weight

list of medications: venlafaxine + bupropion

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (yes)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (yes)	Nausea or emesis

If y then --> give BP meds (not high at baseline, so this would not make sense), **avoid ketamine** and consider propofol and fentanyl

If y then --> Add zofran

Cocktail: Tylenol+ Zofran+ etomidate + succinylcholine

No complications

CASE 5, TREATMENT 8

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. Since initiation of treatment, patient has resumed PO intake, changed psychiatric medications in favor of sympathomimetic effects and has experienced improvement in energy.

Psychiatric and genetic factors no changes

Medical Factors no changes

62 resting HR

110 systolic

65 diastolic

65kg weight

list of medications: venlafaxine + bupropion

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (yes)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (yes)	Nausea or emesis

If y then --> give BP meds (not high at baseline, so this would not make sense), **avoid ketamine** and consider propofol and fentanyl

If y then --> Add zofran

Cocktail: Tylenol+ Zofran+ etomidate + succinylcholine

No complications

CASE 5, TREATMENT 9

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. No complications in prior treatment and she is currently being treated with methohexital and ketamine with Tylenol premedication and she has been tolerating increase of power. Seizure has been shortening, but still adequate.

Psychiatric and genetic factors no changes

Medical Factors no changes

62 resting HR

110 systolic

65 diastolic

65kg weight

list of medications: venlafaxine + bupropion

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Tylenol+ Zofran+ etomidate + succinylcholine

No complications

CASE 5, TREATMENT 10

Patient is a 32 yo female outpatient with history of postpartum depression and PTSD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 62 and BP 110/65 with no known arrhythmias weighing 62kg. No complications in prior treatment and she is currently being treated with methohexital and ketamine with Tylenol premedication and she has been tolerating increase of power. Seizure has been shortening, but still adequate.

Psychiatric and genetic factors no changes

Medical Factors no changes

62 resting HR

110 systolic

65 diastolic

65kg weight

list of medications: venlafaxine + bupropion

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Cocktail: Tylenol+ Zofran+ etomidate + succinylcholine

No complications and patient completed index series after plateauing.

CASE 6, TREATMENT 1

Patient is a 45 yo female outpatient with history of PTSD and MDD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 76 and BP 130/75 with no known arrhythmias weighing 62kg. Presenting for first treatment with DOP negative pregnancy test.

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Dx bipolarity or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Dx chronic pain: headache
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

By default: methohexital and start RUL

If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
---------------	--

Medical Factors

45 Age

76 resting HR

130 systolic

75 diastolic

62kg weight

list of medications: ropinirole, lisinopril, atorvastatin

Prior Complications (Skip if 1st ever treatment)

IV Dosing of ECT anesthetics

	minimum	maximum
methohexital (1-1.5mg/kg)	62	93

Cocktail: [tylenol+methohexital+succinylcholine](#)

Complicated by severe migraine, nausea, and sinus tachycardia.

CASE 6, TREATMENT 2

Patient is a 45 yo female outpatient with history of PTSD and MDD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 76 and BP 130/75 with no known arrhythmias weighing 62kg.

Medical Factors no changes

45 Age

76 resting HR

130 systolic

75 diastolic

62kg weight

list of medications: liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (yes)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol , or consider rocuronium
---------------	--

If y then -->	Add zofran
---------------	------------

Cocktail: [tylenol+toradol+zofran+methohexital+succinylcholine](#)

[Complicated by severe migraine and tachycardia.](#)

CASE 6, TREATMENT 3

Patient is a 45 yo female outpatient with history of PTSD and MDD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 76 and BP 130/75 with no known arrhythmias weighing 62kg.

Medical Factors no changes

45 Age

76 resting HR

130 systolic

75 diastolic

62kg weight

list of medications: liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (no)	Nausea or emesis

If y then -->	Add premedication (Tylenol , Toradol , opioids (remi or dilaudid), or consider rocuronium
---------------	--

Cocktail: tylenol+toradol+zofran+methohexital+succinylcholine+ dilaudid

Complicated by severe migraine and tachycardia.

CASE 6, TREATMENT 4

Patient is a 45 yo female outpatient with history of PTSD and MDD without behavioral issues (but no benzos or other seizure medications or delirium) with a resting HR 76 and BP 130/75 with no known arrhythmias weighing 62kg. Patient with continued migraines despite optimization of reasonable PACU pain management.

Medical Factors no changes

45 Age

76 resting HR

130 systolic

75 diastolic

62kg weight

list of medications: liothyronine 50mcg, Lmethylfolate 15mg, omeprazole 40mg

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (no)	Nausea or emesis

If y then -->	Keep basic premedication (removed dilaudid) and consider rocuronium
---------------	---

Cocktail: tylenol+toradol+zofran+methohexital+succinylcholine+ rocuronium

No complications.

CASE 7, TREATMENT 1

A 54-year-old female outpatient with a history of treatment-resistant major depressive disorder (MDD), PTSD, and generalized anxiety disorder (GAD) presenting for her first ECT treatment with significant anxiety and nausea. She has a history of hysterectomy and chronic back pain and is currently on bupropion, omeprazole, and Ozempic (holding 2 weeks prior to procedure). No history of bipolarity, impulsive violence, seizure medications, or neurocognitive disorder. No known arrhythmias. Resting vitals show HR 102 bpm and BP 145/95. Weight is 70kg.

Age: 54 Weight (kg): 70

Resting HR: High (HR > 100) Resting Systolic BP: High (SBP > 140) Resting Diastolic BP: High (DBP > 90) pharm: bupropion, omeprazole, ozempic

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Prior reemergence delirium or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Current sx's: headache, nausea , bradycardia
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

By default: methohexital and start RUL

	Add premedication (Tylenol, Toradol or other NSAIDs) or consider
If y then -->	rocuronium
If y then -->	Add zofran

Rationale: Given the patient's elevated heart rate and blood pressure, ketamine and volatile anesthetics (sevoflurane, isoflurane) are discouraged due to their potential to exacerbate hypertension and tachycardia. Current exacerbation of chronic pain warrants NSAID premedication. Precedex could be a good agent for anxiety and possible agitation/sympathetic tone with high HR, in addition to gold standard methohexital. Zofran chosen due to subset of nausea under current sx's.

Cocktail: Zofran+ Tylenol+ methohexital + precedex + succinylcholine

Complications: headache

CASE 7, TREATMENT 2

54-year-old female outpatient with MDD and GAD, chronic back pain, and elevated resting HR and BP. After her first ECT treatment, No reemergence delirium, nausea/emesis, prolonged or inadequate seizure noted. Headache complicated recovery. No changes in medication or psychiatric profile.

Age: 54 Weight (kg): 70

Resting HR: High Resting Systolic BP: High Resting Diastolic BP: High

pharm: bupropion, omeprazole, ozempic

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (no)	Nausea or emesis

Rationale: Only complication was persistent headache and so Toradol added on for additional relief, otherwise regimen unchanged.

Cocktail: Zofran+ Tylenol+ Toradol+ methohexital + precedex + succinylcholine

Complications: none

CASE 8 TREATMENT 1

A 45-year-old male outpatient with a history of PTSD and moderate MDD, presenting for his first ECT treatment. He reports chronic neck pain and frequent headaches, but no history of bipolarity, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Vitals show resting HR of 105 bpm, BP 150/95 mmHg, and weight of 85 kg. No known arrhythmias or fractures.

Age: 45 Weight (kg): 85

pharm: sertraline 200mg

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Prior reemergence delirium or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Current pain: headache
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

By default: methohexital and start RUL

If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
---------------	--

Rationale: The default pathway includes methohexital and given active chronic pain, NSAID premedication is recommended for chronic pain and headache. Methohexital is selected as the induction agent due to its favorable seizure profile and absence of contraindications. As the patient insists Tylenol does nothing for him and he is currently experiencing a mild- moderate headache, we discuss and he agrees to Toradol.

Cocktail: Methohexital + Toradol + succinylcholine

Complicated tachyarrhythmia and nausea/emesis post-procedure

CASE 8, TREATMENT 2

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. After his first ECT treatment with methohexital, NSAID premedication, and precedex, he experienced tachyarrhythmia and nausea/emesis post-procedure. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 85

Resting HR: 75 High Resting Systolic BP: 145 Resting Diastolic BP: 88

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (yes)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (yes)	Nausea or emesis

If y then -->	avoid ketamine or tachycardia inducing anesthetics, consider propofol
If y then -->	Add zofran

pharm: sertraline 200mg

Rationale: The presence of tachyarrhythmia necessitates avoiding ketamine and other agents that may exacerbate heart rate and propofol would be the preferred agent. NSAIDs are continued for chronic pain as this was stable during the treatment. Zofran is added to address nausea/emesis.

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 3

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 85

Resting HR: 74 High Resting Systolic BP: 140 Resting Diastolic BP: 86

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

pharm: sertraline 200mg

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 4

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. He notes he lifted weights for the first time in months over the weekend. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 86

Resting HR: 68 High Resting Systolic BP: 142 Resting Diastolic BP: 84

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

pharm: sertraline 200mg

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 5

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 86

Resting HR: 68 High Resting Systolic BP: 142 Resting Diastolic BP: 84

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

pharm: sertraline 200mg

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 6

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. He notes increase in appetite. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 86

Resting HR: 68 High Resting Systolic BP: 142 Resting Diastolic BP: 84

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

pharm: sertraline 200mg

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complications: short seizure even with hyperventilation and retreat

CASE 8, TREATMENT 7

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 86

Resting HR: 68 High Resting Systolic BP: 142 Resting Diastolic BP: 84

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

If y then -- >	Change to etomidate/ add stimulants (less desirable due to AE nausea, tachycardia), Increase power , RUL-> bilateral , increase hyperventilation
-------------------	--

pharm: sertraline 200mg

Rationale: keep anesthetics and switch to bilateral

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 8

Patient Summary: 45-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 45 Weight (kg): 86

Resting HR: 68 High Resting Systolic BP: 142 Resting Diastolic BP: 84

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

pharm: sertraline 200mg

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 9

Patient Summary: 46-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications. Patient notes having started a beta blocker since the last visit.

Age: 46 Weight (kg): 84

Resting HR: 65 High Resting Systolic BP: 132 Resting Diastolic BP: 74

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

pharm: sertraline 200mg

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 10

Patient Summary: 46-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 46 Weight (kg): 84

Resting HR: 65 High Resting Systolic BP: 132 Resting Diastolic BP: 74

pharm: sertraline

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complications means we continue prior regimen.

Cocktail: propofol + Toradol + succinylcholine

Complication: inadequate seizure

CASE 8, TREATMENT 11

Patient Summary: 46-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 46 Weight (kg): 84

Resting HR: 65 High Resting Systolic BP: 132 Resting Diastolic BP: 74

pharm: sertraline

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

If y then -- >	Change to etomidate/add stimulants (less desirable due to AE nausea, tachycardia), Increase power, RUL > bilateral, increase hyperventilation
-------------------	---

Rationale: Since power and laterality are fully optimized, we reconsider the risks vs benefits of changing to etomidate or adding stimulants. In this patient's case, BP and HR have improved at baseline and so tolerability of these stimulating anesthetics may have improved. Will transition back to methohexital, transition to etomidate would also be fair but this provider tends to go with known effects of methohexital with plan to prevent rather than testing patient specific tolerability with etomidate.

Cocktail: methohexital + Toradol + succinylcholine

Complications: none

CASE 8, TREATMENT 12

Patient Summary: 46-year-old male outpatient with PTSD and moderate MDD, chronic neck pain, and frequent headaches. No reemergence delirium, prolonged or inadequate seizure, or bradyarrhythmia noted. No changes in psychiatric profile or medications.

Age: 46 Weight (kg): 84

Resting HR: 65 High Resting Systolic BP: 132 Resting Diastolic BP: 74

pharm: sertraline

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

If y then -- >	Change to etomidate/add stimulants (less desirable due to AE nausea, tachycardia), Increase power, RUL > bilateral, increase hyperventilation
-------------------	---

Rationale: Since power and laterality are fully optimized, we reconsider the risks vs benefits of changing to etomidate or adding stimulants. In this patient's case, BP and HR have improved at baseline and so tolerability of these stimulating anesthetics may have improved. Will transition back to methohexital, transition to etomidate would also be fair but this provider tends to go with known effects of methohexital with plan to prevent rather than testing patient specific tolerability with etomidate.

Cocktail: methohexital + Toradol + succinylcholine

Complications: none

CASE 9, TREATMENT 1

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency, presenting for her first ECT treatment. She also has chronic osteoarthritis-related pain and no history of unstable vertebral fracture. She is currently taking bupropion and omeprazole, but is not on benzodiazepines or seizure medications. No known arrhythmias. Currently bed bound and not eating.

HR of 58 bpm, BP 95/60 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Psychiatric factors most important for the patient case
binary (no)	prior reemergence or hx violence
binary (no)	benzos
binary (yes)	on seizure medications
binary (yes)	current pain
binary (no)	current nausea or emesis
binary (yes)	logistics w. high risk for cognitive issues

	Consider methohexital +/- precedex (not a good idea because of pulse, but can consider if needed later for agitation) AND Avoid ketamine, Sevoflurane, Isoflurane, and midazolam, propofol (may work for some)
If y then -->	Avoid propofol
If y then -->	Add premedication (Tylenol , Toradol or other NSAIDs) or consider rocuronium

Rationale: This patient presents with multiple risk factors: bipolarity, neurocognitive impairment, chronic pain, and low resting vitals. These warrant avoiding ketamine, propofol, and volatile anesthetics. Methohexital is preferred due to bupropion use. Precedex or sufentanyl are used with caution for sedation and cognitive protection, but the patient has low resting HR and BP. NSAIDs are recommended for pain. Started bilateral for more robust response given severity.

Cocktail: Methohexital + Toradol + succinylcholine

Complicated by tachyarrhythmia, agitation reemerging, nausea/emesis post-procedure

CASE 9, TREATMENT 2

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. Since last treatment, she has started speaking short phrases with staff and eating some bites of food.

HR of 58 bpm, BP 95/60 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors/ Prior complications
binary (yes)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (yes)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (yes)	nausea or emesis
binary (yes)	current unstable fracture

If y then -->	avoid ketamine or tachycardia inducing anesthetics, consider propofol	
If y then -->		Add zofran
	Start RUL, prioritize Precedex or sufentanyl and Avoid propofol, sevoflurane,	
If y then -->	isoflurane, ketamine, and midazolam	
If y then -->	Use rocuronium for paralytic; IF ROC THEN ALSO ADD SUGAMMADEX	

Rationale: Based on above, use of propofol would assist with tachycardia and precedex addition would likely assist with any paradoxical agitation that may theoretically occur with use of propofol. Due to prolonged immobility and new diagnosis of fracture, safer to go with roc.

Cocktail: Toradol+ Zofran+ propofol+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 3

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications.

HR of 58 bpm, BP 95/60 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes

Cocktail: Toradol+ Zofran+ propofol+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 4

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. No known arrhythmias. Since last treatment, she has started speaking short phrases with staff and eating some bites of food.

HR of 58 bpm, BP 100/70 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes

Cocktail: Toradol+ Zofran+ propofol+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 5

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. No known arrhythmias. Sleep study showed OSA and patient was started on CPAP

HR of 58 bpm, BP 110/70 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes but required retreatment with hyperventilation.

Cocktail: Toradol+ Zofran+ propofol+ precedex + rocuronium + sugammadex

Complication: Insufficient seizure

CASE 9, TREATMENT 6

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. Tolerating CPAP well

HR of 55 bpm, BP 100/65 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Switch to etomidate for better seizure target as there are still 6 treatments anticipated.

Cocktail: Toradol+ Zofran+ etomidate+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 7

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. No known arrhythmias. Since last treatment, she has started speaking short phrases with staff and eating some bites of food.

HR of 55 bpm, BP 100/65 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes

Cocktail: Toradol+ Zofran+ etomidate+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 8

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. No known arrhythmias. Since last treatment, she has started speaking short phrases with staff and eating some bites of food.

HR of 55 bpm, BP 100/65 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes

Cocktail: Toradol+ Zofran+ etomidate+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 9

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. Patient plateaued.

HR of 55 bpm, BP 100/65 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes

Cocktail: Toradol+ Zofran+ etomidate+ precedex + rocuronium + sugammadex

Complication: None

CASE 9, TREATMENT 10

60-year-old female outpatient with a history of late-onset bipolar disorder and mild neurocognitive impairment and severe vitamin D deficiency. She also has chronic osteoarthritis-related pain and in the interim has been diagnosed with hairline vertebral fracture. Not on benzodiazepines, continues Lamictal seizure medications. Patient plateaued.

HR of 55 bpm, BP 100/65 mmHg, and weight of 68 kg.

pharm: mirtazapine, lamictal, trazodone, atorvastatin, lisinopril, tylenol

Age: 60 Weight (kg): 68

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no changes

Cocktail: Toradol+ Zofran+ etomidate+ precedex + rocuronium + sugammadex

Complication: None

CASE 10 TREATMENT 1

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 79

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (no)	Prior reemergence delirium or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (yes)	Current pain: headache
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

By default: methohexital and start RUL

If y then -->	Add premedication (Tylenol, Toradol or other NSAIDs)
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Rationale: The default pathway includes methohexital and given active chronic pain, NSAID premedication is recommended for chronic pain and headache. Methohexital is selected as the induction agent due to its favorable seizure profile and absence of contraindications.

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 2

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 79

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 3

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 4

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: Inadequate seizure

CASE 10 TREATMENT 5

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: change to bilateral

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 6

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 7

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures. Response plateaued.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 8

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 10 TREATMENT 9

80-year-old male outpatient with a history of MDD and grief. He reports chronic neck pain and frequent headaches, but no history of mania, impulsive violence, neurocognitive disorder, or seizure medications. He is not on benzodiazepines. Hx Afib, no fractures.

Age: 80 Weight (kg): 85

pharm: sertraline, metoprolol, prazosin, gabapentin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: Methohexital + Toradol + succinylcholine

Complications: None

CASE 11 TREATMENT 1

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Psychiatric factors most important for the patient case

Input types	Psychiatric factors most important for the patient case
binary (yes)	Prior reemergence delirium or hx impulsive violence
binary (yes)	On benzodiazepines
binary (no)	On seizure medications
binary (no)	Current pain: headache
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder

Rationale: Start with most sedating anesthetics as patient has numerous attempts at harming staff. Therefore starting with propofol. A practical rule of thumb is that the **flumazenil dose (mg) $\approx$ 0.1 $\times$ lorazepam-equivalent dose (mg)**, so 2–5 mg of lorazepam would need about 0.2–0.5 mg of flumazenil for partial reversal. Because flumazenil's half-life is much shorter, patients may become sedated shortly after administration again, and abrupt reversal in chronic users can trigger withdrawal or seizures. Proactively treating pain can also prevent agitation downstream.

Cocktail: flumazenil + Toradol + tylenol + propofol + succinylcholine

Complicated by agitation/reemergence (failed midazolam, propofol max dose reached, but Ativan worked quickly)

CASE 11 TREATMENT 2

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (yes)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Keep flumazenil, Toradol, Tylenol, propofol

Cocktail: flumazenil + Toradol + tylenol + propofol + succinylcholine + Ativan (worked empirically)

Complicated by agitation/reemergence but grimacing and localizing to back was suspected clinically.

CASE 11 TREATMENT 3

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient in chart has been more responsive and alluding to pain and is visibly shifting around to get comfortable.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (yes)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

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If y then -->	Start RUL , prioritize Precedex (would not address clinical pain but a valid option) or su/fentanyl and Avoid propofol, sevoflurane, isoflurane, ketamine, and midazolam
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Rationale: Keep flumazenil, Toradol, Tylenol, propofol and add precedex

Cocktail: flumazenil + Toradol + tylenol + propofol + succinylcholine + Ativan+ precedex

No complications

CASE 11 TREATMENT 4

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient in chart has been more responsive and alluding to pain and is visibly shifting around to get comfortable.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complication no change

Cocktail: flumazenil + Toradol + tylenol + propofol + succinylcholine + Ativan+ precedex

No complications

CASE 11 TREATMENT 5

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient in chart has been more responsive and alluding to pain and is visibly shifting around to get comfortable.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complication no change

Cocktail: flumazenil + Toradol + tylenol + propofol + succinylcholine + Ativan+ precedex

Complicated by insufficient seizure

CASE 11 TREATMENT 6

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient can identify treatment members and respond appropriately. Still dazed.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 88 Resting Systolic BP: 150 Resting Diastolic BP: 85

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Propofol should be changed to methohexital (etomidate avoided given its nausea and stimulation during reemergence)

Cocktail: flumazenil + Toradol + tylenol + methohexital + succinylcholine + Ativan+ precedex

Complicated by tachyarrhythmia

CASE 11 TREATMENT 7

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient is back to baseline for one on one interactions but still has significant psychomotor retardation. PO intake improving.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 70 Resting Systolic BP: 140 Resting Diastolic BP: 75

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (yes)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Move precedex up to be given with propofol for effect on HR

Cocktail: flumazenil + Toradol + tylenol + precedex + propofol + succinylcholine + Ativan

No complications

CASE 11 TREATMENT 8

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient is back to baseline for one on one interactions but still has significant psychomotor retardation. PO intake improving.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 70 Resting Systolic BP: 140 Resting Diastolic BP: 75

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complication, no changes.

Cocktail: flumazenil + Toradol + tylenol + precedex + propofol + succinylcholine + Ativan

No complications

CASE 11 TREATMENT 9

A 31-year-old male outpatient with a history of MDD and polysubstance use (spice, cocaine, alcohol, cannabis). Mute and catatonic with no report of discomfort and very minimal PO intake. Unclear history of mania, agitation in ICU, no neurocognitive disorder, on Ativan 2mg q4h but no other seizure medications. No known arrhythmias or fractures. Patient is back to baseline and working with therapy. Interested in tapering and focusing on medications and sobriety.

Age: 31 Weight (kg): 100

pharm: Ativan, Thorazine, olanzapine, memantine, prazosin

Resting HR: 70 Resting Systolic BP: 140 Resting Diastolic BP: 75

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complication, no changes.

Cocktail: flumazenil + Toradol + tylenol + precedex + propofol + succinylcholine + Ativan

No complications

Case 12 Treatment 1

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. She has experienced multiple manic episodes, the most recent occurring six months ago, and has a documented history of violence during these episodes, including two incidents of physical aggression. Her medical history also includes difficult anesthesia reemergence, characterized by prolonged emergence and agitation following a prior orthopedic procedure. She is currently on Lamotrigine 200 mg daily, but is not taking any benzodiazepines, having discontinued them three months ago due to concerns about dependency. The patient has lbp and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 88 bpm and a blood pressure of 142/92 mmHg. Her weight is 76 kilograms.

Clinical Factor	Recommendation
History of difficult reemergence, violence, or mania	Consider Methohexital ± Precedex . Avoid: Ketamine, Sevoflurane, Isoflurane, Midazolam
On seizure medications (Lamotrigine)	Avoid Propofol
Current pain (lower back, 6/10)	Add premedication: Tylenol or NSAID
Current nausea	No action needed (nausea denied) Start RUL
Neurocognitive issues (MoCA 23/30)	Prioritize: Precedex or Sufentanil Avoid: Propofol, Sevoflurane, Isoflurane, Ketamine, Midazolam
normal resting HR (88 bpm)	
High BP (142/92 mmHg)	Give BP meds Avoid: Ketamine Consider: Propofol and Fentanyl
Unstable fracture (L2 vertebra)	Use: Rocuronium Avoid: Succinylcholine

Cocktail: methohexital + Precedex + Tylenol+ rocuronium + sugammadex with plan for +/- propofol as severe agitation on reemergence may necessitate this but after seizure has been elicited

Complications: The patient did not experience reemergence delirium, hypertensive urgency/emergency, hypotensive shock, bradyarrhythmia, or prolonged seizure. However, she did experience a mild tachyarrhythmia during the recovery phase, which resolved spontaneously without intervention. The seizure duration was adequate for therapeutic purposes. Post-treatment, the patient reported a mild headache, but no nausea or emesis and propofol was not required

Case 12 Treatment 2

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. She tolerated last treatment with exception of tachyarrhythmia and mild but persistent headache and continues on Lamotrigine 200 mg daily. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 85 bpm and a blood pressure of 135/88 mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (yes)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

Rationale: Patient likely to tolerate propofol given absence of reemergence if it is swapped out for methohexital. The fact that the patient experienced adequate seizure indicates seizure threshold is less of a problematic factor at this point. We can always switch back to methohexital if this becomes a problem over the course of treatment. Anticipate that this change will need to happen as seizure threshold increases with each treatment.

Cocktail: propofol + Precedex + Tylenol+ rocuronium + sugammadex

No complications

Case 12 Treatment 3

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 85 bpm and a blood pressure of 135/88 mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change.

Cocktail: propofol + Precedex + Tylenol+ rocuronium + sugammadex

No complications

Case 12 Treatment 4

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 85 bpm and a blood pressure of 135/88 mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change

Cocktail: propofol + Precedex + Tylenol+ rocuronium + sugammadex

No complications, initially inadequate seizure, but patient was transitioned from RUL to bilateral as more robust response desired and patient without memory complaints or reemergence.

Case 12 Treatment 5

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 85 bpm and a blood pressure of 135/88 mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications no change

Cocktail: propofol + Precedex + Tylenol+ rocuronium + sugammadex

Complications: none

Case 12 Treatment 6

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 85 bpm and a blood pressure of 135/88 mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications no change

Cocktail: propofol + Precedex + Tylenol+ rocuronium + sugammadex

Complications: none

Case 12 Treatment 7

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 85 bpm and a blood pressure of 135/88 mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications no change

Cocktail: propofol + Precedex + Tylenol+ rocuronium + sugammadex

Complications: inadequate seizure. In anticipation of needing to transition to methohexital, we encourage patient to follow up with PCP to initiate BP medication.

Case 12 Treatment 8

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily and initiated on Lisinopril by PCP. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 78 bpm and a blood pressure of 130/75mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Change back to methohexital and anticipate sympathetic storming (draw up labetalol and/or esmolol)

Cocktail: methohexital + Precedex + Tylenol+ rocuronium + sugammadex

Complications: none

Case 12 Treatment 9

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily and initiated on Lisinopril by PCP. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 78 bpm and a blood pressure of 130/75mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications no change

Cocktail: methohexital + Precedex + Tylenol+ rocuronium + sugammadex

Complications: none

Case 12 Treatment 10

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily and initiated on Lisinopril by PCP. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 78 bpm and a blood pressure of 130/75mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change and patient tapering

Cocktail: methohexital + Precedex + Tylenol+ rocuronium + sugammadex

Complications: none

Case 12 Treatment 11

A 65-year-old female with a history of Bipolar I Disorder presents with mild cognitive impairment, evidenced by a MoCA score of 23 out of 30. Continues on Lamotrigine 200 mg daily and initiated on Lisinopril by PCP. She continues to experience lower back pain and unstable L2 vertebral fracture. She denies any current nausea. Her vital signs include a resting heart rate of 78 bpm and a blood pressure of 130/75mmHg. Her weight is 76 kilograms.

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change and last of taper

Cocktail: methohexital + Precedex + Tylenol+ rocuronium + sugammadex

Complications: none

Case 13 Treatment 1

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms.

Input types	Psychiatric factors most important for the patient case
binary (no)	Dx bipolarity or hx impulsive violence
binary (no)	On benzodiazepines
binary (no)	On seizure medications
binary (no)	Current chronic pain
binary (no)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder
Binary (yes)	Nausea

Rationale: The patient with a lower seizure threshold than most, and being younger and therefore less likely to have factors increasing seizure threshold, and having nausea would encourage this provider to use propofol for the first treatment as this is likely to have a more comfortable reemergence; however, many patients are still likely to respond well to methohexital and precedex as that is the option when the strictest of caution is to be taken with the ECT regimen. Use of propofol in our case may also spare the need for addition of Zofran from a polypharmacy standpoint.

Cocktail: propofol + succinylcholine

Complication: Following the initial treatment, the patient did not experience reemergence delirium, hypertensive urgency/emergency, hypotensive shock, or tachyarrhythmia but he did experience bradycardia to the low 40s. There were no prolonged or inadequate seizures observed during the session. The patient also reported no headache or nausea/emesis post-treatment.

Case 13 Treatment 2

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (yes)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: In the case of this young patient, the bradyarrhythmia happened during parasympathetic response to sympathetic surging. Plan to add glycopyrrholate to create floor for pulse.

Cocktail: glycopyrrholate+ propofol + succinylcholine

Complication: Endorses nausea after treatment

Case 13 Treatment 3

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (yes)	Nausea or emesis

Rationale: No complications, no change

Cocktail: glycopyrrholate+ Zofran+ propofol + succinylcholine

Complication: None

Case 13 Treatment 4

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complications, no change

Cocktail: glycopyrrholate+ Zofran+ propofol + succinylcholine

Complication: None

Case 13 Treatment 5

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 100 kilograms. He notes that he has been getting lost in familiar places and does not remember why he started treatments. His mother has been arriving with him to appointments.

Prior Complications:

Input types	Subsequent treatment Factors
binary (yes)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: Given significance of cognitive dulling, will discuss change of anesthetic as patient is on higher end of propofol dosing due to weight and is approaching higher power with treatments as well.

Cocktail: glycopyrrholate+ Zofran+ methohexital+ precedex+ succinylcholine

Complication: None

Case 13 Treatment 6

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms. His mother notes that there has not been a significant improvement in mentation with switch of anesthetics but there has been improvement in mood and self-injurious behavior.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complications, no change. Switch allowed us to reduce power which allows for more therapeutic treatments on current regimen.

Cocktail: glycopyrrholate+ Zofran+ methohexital+ precedex+ succinylcholine

Complication: None

Case 13 Treatment 7

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms. His mother notes that there has not been a significant improvement in mentation with switch of anesthetics but there has been improvement in mood and self-injurious behavior.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complications, no change. Switch allowed us to reduce power which allows for more therapeutic treatments on current regimen.

Cocktail: glycopyrrolate+ Zofran+ methohexital+ precedex+ succinylcholine

Complication: None

Case 13 Treatment 8

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complications, no change.

Cocktail: glycopyrrholate+ Zofran+ methohexital+ precedex+ succinylcholine

Complication: None

Case 13 Treatment 9

A 22-year-old male presents with Autism Spectrum Disorder and Unspecified Depressive Disorder and PTSD who has previously experienced difficult reemergence from anesthesia as well as has a history of “unspecified seizure-like activity”. Not on antiepileptic medication or benzodiazepines. Currently on abilify and hydroxyzine. He reports no current pain, headache, but endorses nausea. There is no evidence of an unstable fracture. His resting heart rate is 72beats per minute, and his blood pressure is 132/74 mmHg. He weighs 76 kilograms. His mother notes that there has not been a significant improvement in mentation with switch of anesthetics but there has been improvement in mood and self-injurious behavior.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complications, no change. End of course

Cocktail: glycopyrrholate+ Zofran+ methohexital+ precedex+ succinylcholine

Complication: None

Case 14, Treatment 1

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety (discussed need to hold the night before). The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 65 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg

Input types	Psychiatric factors most important for the patient case
binary (yes)	Dx bipolarity or hx impulsive violence
binary (yes)	On benzodiazepines
binary (yes)	On seizure medications
binary (no)	Current chronic pain
binary (yes)	Dx neurocognitive disorder
Binary (no)	Dx fracture or neuromuscular disorder
Binary (yes)	Nausea

Rationale: Given the presence of significant neurological disease and high risk for delirium as well as foreseen challenges with obtaining a seizure, this provider would initially lean towards etomidate to guarantee a seizure, but be wary because of worsening nausea (baseline + secondary to etomidate use). Propofol may be helpful for agitation in context of having held Depakote the night before and morning of; however, this would make it more challenging to achieve a seizure. In light of this, methohexital will ensure that we start with the least amount of polypharmacy and it is not likely to cause nausea, prevent a seizure, or cause agitation.

Cocktail: methohexital + succinylcholine + propofol post procedure (or depakote small dose on awakening propofol would be more common based on availability and Depakote would avoid polypharmacy by replacing what was held prior to procedure)

Complication: Copious secretions limiting visibility and increasing risk of aspiration

Case 14 Treatment 2

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg. Copious secretions on last visit.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (yes)	Secretions

Rationale: need glycopyrrolate to manage airway better, low pulse indicates this may be helpful for prevention of bradyarrhythmia as well

Cocktail: glycopyrrolate + methohexital + succinylcholine , anesthesia tech forgot to give propofol but patient awakened without reemergence

Complications: None

Case 14 Treatment 3

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: No complication no change

Cocktail: glycopyrrolate + methohexital + succinylcholine

Complications: Reemergence delirium noted which ceased with propofol 30mg

Case 14 Treatment 4

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: No complication no change

Cocktail: glycopyrrolate + methohexital + succinylcholine + propofol 30mg post procedure

Complications: None

Case 14 Treatment 5

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: No complication no change

Cocktail: glycopyrrolate + methohexital + succinylcholine + propofol 30mg post procedure

Complications: None

Case 14 Treatment 6

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: No complication no change

Cocktail: glycopyrrolate + methohexital + succinylcholine + propofol 30mg post procedure

Complications: None

Case 14 Treatment 7

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: No complication no change

Cocktail: glycopyrrolate + methohexital + succinylcholine + propofol 30mg post procedure

Complications: None

Case 14 Treatment 8

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: No complication no change

Cocktail: glycopyrrolate + methohexital + succinylcholine + propofol 30mg post procedure

Complications: inadequate seizure

Case 14 Treatment 9

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg. Patient initially had a great response but as of treatment #8, this has plateaued

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: Since etomidate can help obtain seizure and it also increases or is neutral for HR and BP, plan to switch methohexital for etomidate but keep glycopyrrolate. Given significance of nausea and likelihood that etomidate will make it worse, will plan for Zofran as well.

Cocktail: glycopyrrolate+ Zofran+ etomidate+ succinylcholine + propofol 30mg post procedure

Complications: headache

Case 14 Treatment 10

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: Add Tylenol for headaches

Cocktail: glycopyrrolate+ Zofran+ Tylenol+ etomidate+ succinylcholine + propofol 30mg post procedure

Complications: none

Case 14 Treatment 11

A 58-year-old male presents for evaluation prior to his first electroconvulsive therapy (ECT) session. He has a diagnosis of schizoaffective disorder with moderate cognitive impairment (MoCA score: 21/30). He has a documented history of manic episodes, with the most recent occurring approximately 3 months ago, and has exhibited violent behavior during past psychiatric hospitalizations. There is no known history of difficult anesthesia reemergence. He is currently prescribed high doses of valproic acid for mood stabilization and seizure do, and is also taking clonazepam for anxiety. The patient reports no current pain or headache, but does complain of mild nausea. He does not have any unstable fractures. His resting heart rate is 60 bpm, and his blood pressure is 128/78 mmHg on beta blockers. His weight is 100 kg. Here for last treatment which is being tapered.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis
Binary (no)	Secretions

Rationale: Add Tylenol for headaches

Cocktail: glycopyrrolate+ Zofran+ Tylenol+ etomidate+ succinylcholine + propofol 30mg post procedure

Complications: none

Case 15 Treatment 1

The patient is a 36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. However, she is experiencing hemodynamic instability, with a resting heart rate of 122 beats per minute and a blood pressure of 88/54 mmHg. Her weight is 58 kilograms.

mal):

Clinical Factor		Recommendation
Prior difficult reemergence	No	Consider methohexital, propofol, or etomidate
History of violence	No	Consider methohexital, propofol, or etomidate
History of mania	No	Consider methohexital, propofol, or etomidate
On seizure medications	No	Propofol is not contraindicated
Current pain	No	No need for Tylenol or NSAID premedication
Current nausea	No	No need for Zofran
Neurocognitive issues	No	No need to prioritize RUL or avoid specific agents
Resting heart rate	High (122 bpm)	Avoid ketamine and tachycardia-inducing agents; consider propofol
Resting blood pressure	Low (88/54 mmHg)	Favor ketamine; avoid hypotension-inducing agents; consider ideal pressor support
Unstable fracture	No	Use succinylcholine

Rationale: Of particular concern prior to starting this case is the patient's low blood pressure. Starting the IV and initiating some fluid resuscitation and labs prior to procedure would be preferred as this would likely suggest clinically that the patient is significantly dehydrated which often leads to electrolyte abnormalities. To keep things simple, we can start with methohexital, in case the low blood pressure is not corrected by our initial interventions, it is BP neutral and although etomidate may increase BP (which may be beneficial in some ways) it can also be more prone to heart rate/rhythm abnormalities and as such may not be as tolerable. As safety is always our primary aim especially during the first procedure where there are more unknown variables.

Cocktail: methohexital + succinylcholine

Complications: tachyarrhythmia and headache during recovery at which time labs from earlier resulted with low Mg. Improved with small dose of esmolol + Mg.

Case 15 Treatment 2

The patient is a 36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 85 beats per minute and a blood pressure of 95/65 mmHg. Her weight is 58 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (yes)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (yes)	headache
binary (no)	nausea or emesis

Rationale: Given known etiology of tachyarrhythmia, add Mg to the regimen. Given headache, will add Tylenol. Regarding magnesium, this can raise the seizure threshold and so this effect will have to be better understood in the future.

Cocktail: Mg+ Tylenol+ methohexital + succinylcholine (robust hyperventilations)

Complications: None.

Case 15 Treatment 3

The patient is a 36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 85 beats per minute and a blood pressure of 95/65 mmHg. Her weight is 58 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change.

Cocktail: Mg+ Tylenol+ methohexital + succinylcholine (robust hyperventilations)

Complications: None.

Case 15 Treatment 4

The patient is a 36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 80 beats per minute and a blood pressure of 95/65 mmHg. Her weight is 60 kilograms. She admits that she was eating 0-1 meals per day and has now increased intake by almost 200%.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications no change.

Cocktail: Mg+ Tylenol+ methohexital + succinylcholine (robust hyperventilations)

Complications: RUL 2nd treatment was 25 seconds but limited quality. Plan to transition to bilateral.

Case 15 Treatment 5

The patient is a 36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 85 beats per minute and a blood pressure of 95/65 mmHg. Her weight is 60 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (yes)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Held Mg as patients PO intake has improved and this is likely interfering with eliciting seizures. Continue Tylenol. Will also transition to bilateral.

Cocktail: Tylenol+ methohexital + succinylcholine

Complications: None.

Case 15 Treatment 6

The patient is a 36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 75 beats per minute and a blood pressure of 100/70 mmHg. Her weight is 60 kilograms. She notes she has been venturing out into coffee shops and spending some more time with friends. She does note some mild self-limited episodic disorientation but states this does not concern her.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change.

Cocktail: Tylenol+ methohexital + succinylcholine

Complications: None.

Case 15 Treatment 7

36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 75 beats per minute and a blood pressure of 100/70 mmHg. Her weight is 60 kilograms. She reports having initiated therapy yesterday for PTSD and had less sleep than the night before.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change.

Cocktail: Tylenol+ methohexital + succinylcholine

Complications: Transient asystole due to failure to elicit sufficient seizure. Patient admits after procedure that due to anxiety after therapy appointment yesterday her friend gave her two Klonopin but didn't think it was important to disclose. She was provided atropine which resulted in immediate reversion back to normal sinus rhythm.

Case 15 Treatment 8

36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 75 beats per minute and a blood pressure of 100/70 mmHg. Her weight is 60 kilograms. She is apologetic for the prior session and notes that she has not had any OTC or sedative medications. UDS is done as well which confirms absence of drugs of abuse and prescriptions.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (yes)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: Given patient's increasing seizure threshold will transition to etomidate.

Cocktail: Tylenol+ etomidate + succinylcholine (robust hyperventilations)

Complications: None.

Case 15 Treatment 9

36-year-old female with normal cognitive function. She has no history of manic episodes or violent behavior, and previous anesthesia reemergence has been uneventful. She is not currently on antiepileptic or benzodiazepine medications. She denies any current pain, headache, or nausea, and there is no evidence of an unstable fracture. Resting heart rate of 75 beats per minute and a blood pressure of 100/70 mmHg. Her weight is 60 kilograms. Patient notes she is mostly back to her baseline and notes that working on PTSD as her next goal.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Em/Urgency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
binary (no)	headache
binary (no)	nausea or emesis

Rationale: No complications, no change.

Cocktail: Tylenol+ etomidate + succinylcholine (robust hyperventilations)

Complications: None.

Case 16 Treatment 1

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Rationale:

Clinical Factor		Recommendation
Prior difficult reemergence	No	Consider methohexital, propofol, or etomidate
History of violence	No	Consider methohexital, propofol, or etomidate
History of mania	No	Consider methohexital, propofol, or etomidate
On antiepileptic medication	Yes	Avoid propofol
On benzodiazepine medication	No	No specific adjustment
Current pain	No	No need for premedication Tylenol or NSAID
Current nausea	No	No need for Zofran
Current neurocognitive issues	No	No special considerations
Resting heart rate	Normal (72 bpm)	No need for glycopyrrolate or atropine; no tachycardia concerns
Resting systolic/diastolic BP	Normal (118/76 mmHg)	No need for BP meds or pressors; no need to avoid/favor specific agents based on BP
Current unstable fracture	No	Use succinylcholine

Cocktail: methohexital + succinylcholine

Complications: Following the initial treatment, the patient experienced transient bradycardia that did not require intervention. She did not exhibit reemergence delirium, hypertensive urgency or emergency, bradyarrhythmia, tachyarrhythmia, prolonged seizure, or inadequate seizure activity. Additionally, she did not report headache or nausea/emesis.

Case 16 Treatment 2

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache,

or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (yes)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: Given bradycardia will add glycopyrrolate.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 3

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure

Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complication, no change.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 4

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complication, no change.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 5

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complication, no change.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 6

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complication, no change.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 7

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complication, no change.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 8

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: No complication, no change.

Cocktail: glycopyrrolate+ methohexital + succinylcholine

Complications: Headache

Case 16 Treatment 9

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (yes)	headache
Binary (no)	Nausea or emesis

Rationale: Given headache will add toradol

Cocktail: Toradol+ glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 16 Treatment 10

A 35-year-old female presents with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is currently taking antiepileptic dose of lamotrigine but is not on any benzodiazepines. She denies experiencing pain, headache, or nausea, and has no unstable fractures. Her resting heart rate is 72 beats per minute, and her blood pressure is 118/76 mmHg. She weighs 64 kilograms. Patient has completely remitted from depressive symptoms.

Prior Complications:

Input types	Subsequent treatment Factors
binary (no)	Reemergence Delirium
binary (no)	Hypertensive Emergency
binary (no)	Hypotensive Shock
binary (no)	Bradyarrhythmia
binary (no)	Tachyarrhythmia
binary (no)	Prolonged seizure
binary (no)	Inadequate seizure
Binary (no)	headache
Binary (no)	Nausea or emesis

Rationale: Given headache will add toradol

Cocktail: Toradol+ glycopyrrolate+ methohexital + succinylcholine

Complications: No complications

Case 17 Treatment 1

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: The patient will be holding her benzodiazepine and receiving flumazenil and therefore ability to elicit a seizure should be possible and in this case, the bigger concern may be agitation from benzo withdrawal. Given high resting pulse and BP, will use propofol. As the benzodiazepine was held last night, there is little concern that the circulating amount will be sufficient to interfere with seizure, if however there is a difficulty eliciting seizure, flumazenil would be the next best option. Zofran for nausea.

Cocktail: Zofran + propofol + succinylcholine

Complication: Reemergence delirium was noted, though limited.

Case 17 Treatment 2

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: Given reemergence delirium, will add precedex as well to spare seizure threshold but provide coverage for agitation.

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: None

Case 17 Treatment 3

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to

nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: No complications, no change

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: None

Case 17 Treatment 4

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: No complications, no change.

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: None

Case 17 Treatment 5

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: No complications, no change

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: transition to bilateral due to difficulty eliciting seizure, although that was effective

Case 17 Treatment 6

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: No complications, no change (for medications), treatment parameters were changed instead

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: None

Case 17 Treatment 7

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: No complications, no change (for medications), treatment parameters were changed instead

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: None

Case 17 Treatment 8

The patient is a 55-year-old female with normal cognitive function and no history of manic episodes, violence, or complications during anesthesia reemergence. She is not on antiepileptic

medication but is currently prescribed benzodiazepines for severe REM sleep behavior disorder that she held last night. She does not report any pain or headache but eventually admits to nausea, and there is no indication of an unstable fracture. Her resting heart rate is 85 beats per minute, and her blood pressure is 140/80 mmHg. She weighs 66 kilograms.

Rationale: No complications, no change (for medications), treatment parameters were changed instead

Cocktail: Zofran + precedex+ propofol + succinylcholine

Complication: Inadequate seizure but patient lost to follow up

Case 18 Treatment 1

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms.

Rationale: gold standard because no treatment modifiers identified

Cocktail: Toradol+ methohexital + succinylcholine

Complications: Following her first treatment, the patient experienced mild tachycardia and hypotension which improved with increased fluid rate during the recovery period. She also reported significant anxiety in the recover bay. She also reported a persistent headache post-procedure. No other complications were observed, there were no signs of reemergence delirium, hypertensive urgency/emergency, hypotensive shock, bradyarrhythmia, prolonged seizure, inadequate seizure activity, or nausea/emesis.

Case 18 Treatment 2

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms.

Rationale: given anxiety, but no reemergence and headache will transition to propofol given its more tolerable side effect profile

Cocktail: Toradol+ propofol + succinylcholine

Complications: None

Case 18 Treatment 3

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but

denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms.

Rationale: no complications, no change

Cocktail: Toradol+ propofol + succinylcholine

Complications: None

Case 18 Treatment 4

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms.

Rationale: no complications, no change

Cocktail: Toradol+ propofol + succinylcholine

Complications: None

Case 18 Treatment 5

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms. Full remission of symptoms and plateau described.

Rationale: no complications, no change

Cocktail: Toradol+ propofol + succinylcholine

Complications: None

Case 18 Treatment 6

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking

antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms.

Rationale: no complications, no change

Cocktail: Toradol+ propofol + succinylcholine

Complications: None

Case 18 Treatment 7

A 23-year-old female presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. She is not currently taking antiepileptic or benzodiazepine medications. The patient reports experiencing a headache but denies any nausea or pain elsewhere. There is no evidence of an unstable fracture. Her resting heart rate is 65 beats per minute, with a blood pressure of 100/80 mmHg, and she weighs 54 kilograms. End of course, no taper required due to Lazarus effect and positive prognosis.

Rationale: no complications, no change

Cocktail: Toradol+ propofol + succinylcholine

Complications: None

Case 19 Treatment 1

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale:

Anesthetic Consideration		
Prior difficult reemergence	No	Consider methohexital, propofol, or etomidate
History of violence	No	Consider methohexital, propofol, or etomidate
History of mania	No	Consider methohexital, propofol, or etomidate

On antiepileptic medication	No	Propofol is not contraindicated
On benzodiazepine medication	No	No specific adjustment needed
Current pain	Yes	+ premedication with Tylenol or NSAID
Current nausea	No	No need for Zofran
Neurocognitive issues	No	No need to prioritize Precedex or avoid specific agents
Resting heart rate	Normal (76 bpm)	No need for glycopyrrolate or atropine
Resting systolic/diastolic BP	Normal (124/78 mmHg)	No need to favor or avoid specific agents based on BP
Unstable fracture	Yes	Use rocuronium for paralytic; avoid succinylcholine

Cocktail: Toradol + Tylenol +methohexital + rocuronium +sugammadex

Complication: Following the initial procedure, the patient experienced a prolonged seizure during the treatment session, requiring extended monitoring and supportive care. He did not exhibit signs of reemergence delirium, hypertensive urgency/emergency, hypotensive shock, bradyarrhythmia, tachyarrhythmia, inadequate seizure activity, headache, or nausea/emesis. The unstable fracture was managed appropriately with rocuronium

Case 19 Treatment 2

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: Given prior prolonged seizure requiring propofol for termination, will change to this as the primary anesthetic

Cocktail: Toradol + Tylenol +propofol + rocuronium +sugammadex

Complication: None

Case 19 Treatment 3

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol + rocuronium +sugammadex

Complication: Patient experienced some reemergence delirium

Case 19 Treatment 4

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: Add precedex for reemergence delirium

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None

Case 19 Treatment 4

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None

Case 19 Treatment 5

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None

Case 19 Treatment 6

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None

Case 19 Treatment 7

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None

Case 19 Treatment 8

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking

antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None

Case 19 Treatment 9

A 57-year-old male presents with normal cognitive function and no history of manic episodes, violent behavior, or complications during anesthesia reemergence. He is not currently taking antiepileptic or benzodiazepine medications. He reports no headache, or nausea at this time but does report significant neck, back, and arm (fracture). However, he has been diagnosed with an unstable fracture requiring orthopedic evaluation. His resting heart rate is 90 beats per minute, and his blood pressure is 140/85 mmHg. He weighs 82 kilograms.

Rationale: No complications, no change

Cocktail: Toradol + Tylenol +propofol +precedex+ rocuronium +sugammadex

Complication: None end of series.

Case 20 Treatment 1

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale:

Anesthetic Consideration		
Prior difficult reemergence	Yes	Consider methohexital ± Precedex; Avoid ketamine, sevoflurane, isoflurane, midazolam; propofol may work for some
History of violence	Yes	Same as above
History of mania	Yes	Same as above
On seizure medications	Yes	Avoid propofol

On benzodiazepines	No	No specific adjustment
Current pain	No	No premedication needed
Current nausea	No	No antiemetic needed
Neurocognitive issues	No	No special adjustment
Resting heart rate	82 bpm (normal)	No HR-related adjustment needed
Resting BP	158/88 mmHg (elevated)	Avoid ketamine; consider propofol and fentanyl; administer BP meds
Unstable fracture	No	Use succinylcholine for paralytic

Cocktail: methohexital + succinylcholine

Complications: None

Case 20 Treatment 2

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no change.

Cocktail: methohexital + succinylcholine

Complications: None

Case 20 Treatment 3

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an

unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no change.

Cocktail: methohexital + succinylcholine

Complications: None

Case 20 Treatment 4

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no change.

Cocktail: methohexital + succinylcholine

Complications: Inadequate seizure in spite of transition in ECT stimulus parameters like transition to bilateral and robust hyperventilation

Case 20 Treatment 5

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: Given inadequate seizure transition to etomidate

Cocktail: etomidate + succinylcholine, required Zofran after

Complications: nausea

Case 20 Treatment 6

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records.

Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: Given nausea, add zofran

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 7

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 8

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 9

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 10

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 11

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 12

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none

Case 20 Treatment 13

The patient is a 67-year-old female with preserved cognitive function. She has a history of manic episodes and prior violent behavior, which has been documented in her psychiatric records. Previous surgical procedures have been complicated by difficult anesthesia reemergence. She is currently taking antiepileptic medication but is not prescribed benzodiazepines. At the time of evaluation, she denies experiencing pain, headache, or nausea. There is no indication of an unstable fracture. Her resting heart rate is 82 beats per minute, and her blood pressure is elevated at 158/88 mmHg. She weighs approximately 68 kilograms.

Rationale: No complications, no changes

Cocktail: Zofran+ etomidate + succinylcholine

Complications: none